

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 9: Some Applications of Trigonometry

Previous year board questions, topic-wise: 5 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 One right-angled triangle

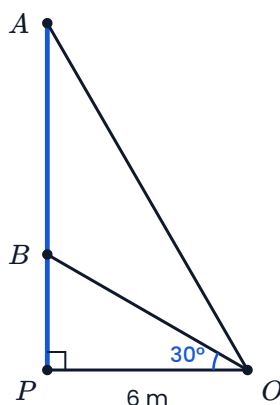
1. A car drives away from the foot of a tower 30 m high. What is the angle of elevation of the top of the tower from the car when it is $10\sqrt{3}$ m from the foot? (a) 30° (b) 45° (c) 90° (d) 60°

CBSE 2026, 1 mark

2. A kite is flying 150 m above the ground. Its string is straight and makes an angle of 30° with the ground. How long is the string? CBSE 2025, 1 mark

2 Two right-angled triangles

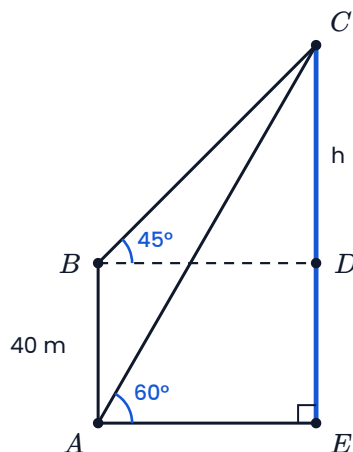
3. A radio tower PA is built in two sections: PB is the lower section and BA the upper one. Wires from a point O on the ground, 6 m from the foot P , hold the tower at B and at A . From O the angle of elevation of B is 30° and that of A is 60° . (i) Find the length of the wire OB . (ii) Find the length of the wire OA . (iii) Find AB , OR find the area of $\triangle OPB$. CBSE 2026, 4 marks



Angle $POB = 30^\circ$ and angle $POA = 60^\circ$

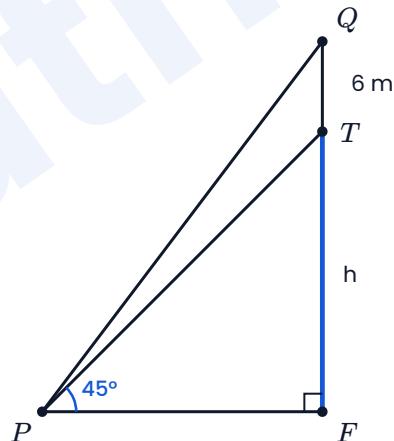
4. From a point A on the ground, the top C of a lighthouse CE is seen at an angle of elevation of 60° . From a deck at B , 40 m directly above A , the angle of elevation of C is 45° . D is the point on CE level with B , and $CD = h$ metres. (i) Write BD in terms of h . (ii) Write BC in terms of h . (iii) Find the height CE , taking $\sqrt{3} = 1.73$, OR find AE if $AC = 100$ m.

CBSE 2025, 4 marks



5. A 6 m pole stands on top of a tower. From a point P on the ground, the top of the pole is seen at an angle of elevation of 60° . From the top of the tower, P is seen at an angle of depression of 45° . Find the height of the tower and the distance of P from its foot. (Take $\sqrt{3} = 1.73$.)

CBSE 2024, 5 marks



FT is the tower and TQ is the pole

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.